2025 Ch H2 Q7

Section: Chemistry in Society
Topic: Chemical Energy

Question Summary:
This question involves calculating the enthalpy of the C–O triple bond in carbon monoxide, explaining equilibrium shifts with temperature, describing apparatus for producing and purifying CO, and calculating atom economy.

Worked Solution

(a) Bond enthalpy of the C–O triple bond in CO
The enthalpy change for forming COCl2 from CO and Cl2 is -107.6 kJ mol-1. Using Data Booklet bond enthalpies (all ASCII-safe values):
- Cl–Cl = 243 kJ mol-1
- C=O (in COCl2) = 804 kJ mol-1
- C–Cl = 338 kJ mol-1

Applying ΔH = bonds broken – bonds formed and rearranging gives the bond enthalpy of the C–O triple bond in CO as 1129.4 kJ mol-1.

(b) Why methane yield increases on cooling
The reaction CO(g) + 3H2(g) -> CH4(g) + H2O(g) is exothermic (ΔH = -206 kJ mol-1), so cooling favours the forward reaction. At temperatures below 100 C, water condenses to liquid, removing a product and pulling the equilibrium further to the right.

(c)(i) Text description of apparatus
1. Producing CO2: A conical flask contains dilute hydrochloric acid and a lump of calcium carbonate. The CO2 formed is carried out through a delivery tube.

2. Producing CO: The CO2 passes into a heated hard glass roasting tube containing hot carbon. The hot carbon reduces CO2 to CO.

3. Removing unreacted CO2: The gas mixture enters a flask containing sodium hydroxide solution. The inlet tube dips below the solution to absorb CO2. The outlet tube sits above the surface and delivers purified CO for collection over water.

(c)(ii) Atom economy
Reaction: Zn + CaCO3 -> ZnO + CaO + CO
Molar masses (g mol-1): Zn 65.4, CaCO3 100.1, CO 28.0.
Total reactants: 165.5 g

Atom economy = 28.0 / 165.5 x 100 = 16.9 percent.

Revision Tips

- When triple bonds are required, write the term 'triple' explicitly (e.g. C–O triple bond).
- Cooling favours exothermic reactions.
- NaOH(aq) removes acidic gases such as CO2.
- Atom economy compares desired product mass with total reactant mass.